

Tutorial:Advanced Rocket Design

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Getting to learn basic rocket science for a space game like Kerbal Space Program can be very important to the success of building rockets that can perform a desired job. In this guide, we will be covering things like calculating the full Delta-V of your ship, explaining how to perform transfer maneuvers, getting Thrust to Weight Ratio, calculating the Peak G-force experienced during a particular burn, also calculating Delta-V needed for a full Hohmann transfer and much more.

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Delta-V

Δv (change in velocity) is the bread and butter of rocket science. It is probably the most important thing to know about your rocket because it determines what your rocket is capable of achieving. In simple terms, Δv is a measure of your craft's ability to affect it's current velocity, something you'll be wanting to do a lot of as you explore the vastness of space.

Among the several things we will explain in this basic tutorial, Δv is most likely the most useful thing you will apply to Kerbal Space Program while building a rocket. To find the Δv of your rocket for a given stage we have to sum up the part masses of every single part of the stage.

- Total mass: m_{total}
- Fuel mass: m_{fuel}
- Dry Mass: $m_{\text{dry}} = m_{\text{total}} - m_{\text{fuel}}$

The calculation only requires the total and dry mass, but it is easier to obtain the dry mass by subtracting the fuel mass from the total mass. Of course other combinations like calculating the total mass and measuring the fuel and dry mass are also possible.

The next important part of this set of calculations is to find your engine's specific impulse. Specific impulse is a measure of how fuel efficient an engine is (the greater the specific impulse, the more fuel efficient it is). For example, the non-vectoring stock engine LV-T30 has a vacuum specific impulse of 300 s. So here, we must apply the Tsiolkovsky rocket equation. More informally known as "The Rocket Equation".

It states:

$$\Delta v = I_{\text{sp}} \cdot \ln \left(\frac{m_{\text{total}}}{m_{\text{dry}}} \right)$$

If the specific impulse is given in seconds it is necessary to multiply this value by $9.82 \frac{\text{m}}{\text{s}^2}$ (see also Terminology about I_{sp}).

We can make sense of this equation by considering Newton's laws of motion. Many people are familiar with his second law, which states that force is the product of mass and acceleration, or $F = m a$. However, since rockets propel themselves by expelling matter in one direction to build velocity in the opposite direction, their mass constantly changes, breaking the requirement of m being a constant in $F = m a$. Fortunately, there is another way of writing Newton's second law; $F = \frac{\Delta p}{\Delta t}$. In this equation, p represents momentum, the product of mass and velocity. The equation then reads as "applying a force results in a change in momentum over a period of time". By rearranging we can obtain $F \Delta t = \Delta p$, telling us that by applying a force for some period of time, we can obtain a change in our momentum! Since this momentum change occurred by throwing matter out behind us to build speed, it is made up of both a Δm and a Δv (we lost mass, and gained velocity). This means that as we burn fuel, our acceleration increases according to $F = m a$, hence the presence of mass in the rocket equation.

The rocket equation also reveals the biggest headache in all of rocket science. By inspecting the equation, we see that there are two ways to increase Δv : Increasing engine efficiency, or increasing total mass to dry mass ratio (think of this in terms of how much fuel you have). One might expect that doubling our carried fuel might double our Δv , but unfortunately this is not the case. We can demonstrate this with an example. Imagine a craft with a dry mass of 200kg, carrying 1000kg of fuel. This gives a total mass of 1200kg and a total to dry mass ratio of 6. By the rocket equation, we multiply our engine impulse by $\ln(6)$, and obtain a certain value. Considering a second identical rocket with double the fuel mass (for a total mass of 2200kg), we find Δv to be engine impulse times $\ln(11)$. By dividing these two results (engine impulse cancels out), we can calculate the increase in Δv , let's see what happens!

$$\frac{\ln(11)}{\ln(6)} = 1.338$$

Uh oh, that's only a 34% increase, but that's the reality of rocket science. Since increased fuel means increased mass, we also get decreased acceleration (and therefore less change in velocity). This effect is captured by the natural log function in the rocket equation, and means that adding fuel to our craft gives exponentially diminishing returns.

To apply the equation, go ahead and sum up your stage's total mass with fuel. Then, go ahead and sum up the mass minus the fuel (this can be done by just adding up the 'dry mass' where given). Input these into the equation in the place of m_{total} and m_{dry} . Following is a quick example, where the surface gravity of Earth $9.81 \frac{\text{m}}{\text{s}^2}$ is used:



Example rocket

Stage 3 (TMI, Mun lander, Return)	
m_{total}	3.72t
m_{dry}	1.72t
I_{sp}	400s
Δv	$3027.0 \frac{\text{m}}{\text{s}}$
Stage 2 (Kerbin orbit insertion)	
m_{total}	7.27t
m_{dry}	5.27t
I_{sp}	300s
Δv	$946.8 \frac{\text{m}}{\text{s}}$
Stage 1 (Ascent):	
m_{total}	38.52t
m_{dry}	14.52t
I_{sp}	350s (estimated due to atmospheric flight)
Δv	$3349.9 \frac{\text{m}}{\text{s}}$
Total	
Δv	$7544.6 \frac{\text{m}}{\text{s}}$

Multiple engines

To calculate the I_{sp} for multiple engines with different I_{sp} values, you need to find total thrust and mass flow:

$$I_{\text{sp}_{\text{avg}}} = \frac{\sum_i^n (\text{thrust}_i)}{\sum_i^n (\dot{m}_i \cdot g_0)} = \frac{\sum_i^n (\text{thrust}_i)}{\sum_i^n \left(\frac{\text{thrust}_i}{I_{\text{sp}_i}} \right)} = \frac{\text{thrust}_1 + \text{thrust}_2 + \dots + \text{thrust}_n}{\text{thrust}_1 \div I_{\text{sp}_1} + \text{thrust}_2 \div I_{\text{sp}_2} + \dots + \text{thrust}_n \div I_{\text{sp}_n}}$$

This will give you the correct I_{sp} to use for your Δv calculation. If all engines are the same, they act as one engine in this calculation so the sums aren't needed.

Calculating transfer maneuvers

The next part of this tutorial is how to perform a transfer maneuver. This kind of action is called a Hohmann Transfer and it requires two burns at opposite points in an orbit. Adding velocity will boost our apoapsis higher. We would then simply wait until we hit our newly established apoapsis and then add more velocity to boost our periapsis to circularize. Or, we could drop our orbit by subtracting velocity by burning retro-grade.

We can also apply some Δv calculations to find out how much thrust we will need to perform this maneuver. We will break this burn up into impulses. For example purposes, we will start at a 100 km orbit and then boost into a 200 km orbit. Both circularized. The formula for the first burn is the following:

$$\Delta v_1 = \sqrt{\frac{\mu}{r_1 + R}} \left(\sqrt{\frac{2(r_2 + R)}{r_1 + r_2 + 2R}} - 1 \right)$$

This is the formula for the final burn in the transfer:

$$\Delta v_2 = \sqrt{\frac{\mu}{r_2 + R}} \left(1 - \sqrt{\frac{2(r_1 + R)}{r_1 + r_2 + 2R}} \right)$$

Where:

- μ = Gravitational parameter of parent body (3530.461 km³/s² for Kerbin).
- r_1 = The altitude of our first orbit (100 km in this case).
- r_2 = The altitude of our second orbit (200 km in this case).
- R = The radius of parent body (600 km in this case).

This formula will give us our velocity for the burn in km/s (multiply by 1000 to convert it into m/s). It's important to make sure that you will have the Δv in the stage to make this burn. Again, you can do that by using the Δv calculations above.

In our case we get a Δv_1 of 73.65 m/s, a Δv_2 of 71.23 m/s and a total Δv of 144.88 m/s.

Calculating fuel flow

Next, we will explain how to calculate fuel flow in mass to see how much fuel a burn uses up in a specific amount of time.

If we know the Δv needed for the burn and the total mass of the rocket before the burn, we can calculate how much fuel is required to complete the burn.

First, we calculate the mass of the rocket after the burn is complete. To do this, we use the Tsiolkovsky Rocket Equation, inputting the initial mass and Δv of the burn. We can then solve the equation for the final mass ("dry mass") after the burn. The difference between these two masses will be used to determine the length of time that is needed to complete the burn.

The equation for mass flow rate of fuel, given I_{sp} and thrust, is:

$$\dot{m} = \frac{\text{thrust}}{I_{sp}}$$

where $\dot{m}$ is the mass flow rate of fuel consumed. Again if the specific impulse is given in seconds it needed to multiplied by 9.81 m·s⁻² (see also Terminology about I_{sp}).

Dividing the difference between initial mass and final mass for the burn by the mass flow rate of fuel, we can determine how many seconds are required.

Usually, when the thrust is in kN and the specific impulse is in m/s the result is in Mg/s (= t/s). As the density of the liquid fuel/oxidizer mixture is 5 Mg/m³ this gives 1/5 m³/s = 2 dm³/s (= l/s).

Orbital velocity

Rather easy is the formula to calculate the orbital velocity of an orbit. This assumes circular orbit or the velocity of a specific point in an orbit. For this, we simply do this calculation:

$$\sqrt{\frac{\mu}{r}}$$

Where:

μ = Gravitational Parameter of parent body. (km³/s²)

r = radius of orbit. (km)

If we input the radius of the orbit in Kilometers, our orbital velocity will come out in Kilometers per second. In a 100 km orbit, our radius will be 700 km. Meaning our velocity will be ~2.2458 kilometers per second (km/s), or 2245.8 m/s.

Delta-v map

A Δv map consists of approximate amounts of Δv needed to get from one place (whether it is on the ground or in space) to another. The Δv values we have for our Δv map are approximate and include a fudge factor (in case we slip up on our piloting). Our map is as follows:

Launch to 100 km Kerbin orbit:	3500 m/s
Trans-Munar Injection:	900 m/s
Landing on the Mun:	1000 m/s
Launch from Mun and return to Kerbin:	1000 m/s
Total Δv :	6400 m/s

If we design our rockets to have 6400 total Δv , and the acceleration of the launch stages are adequate, we can have confidence that our rocket is able to land on the Mun and return to Kerbin. A rocket with a little less Δv can accomplish this goal, but it is less forgiving of less efficient piloting.

Calculate the acceleration

→ See also: [Thrust-to-weight ratio](#)

Calculating the thrust-to-weight ratio is very simple. It is important to know the thrust to weight ratio of your rocket to ensure your rocket will actually liftoff. If your TWR is less than 1, you can bet that you won't make an inch in altitude when starting from the launch pad. The minimum optimal TWR to have for your rocket at launch is 2.2.

To lift off the rocket's thrust need to exceed the gravitational force. The formula for this is simply the thrust of all of your current stage engines divided by the weight of your ship, fully fuelled.

$$F_T > F_G = m \cdot g \Rightarrow TWR = \frac{F_T}{F_G} = \frac{F_T}{m \cdot g} > 1$$

To calculate the acceleration simply use [Newton's second law](#):

$$F = m \cdot a = F_T - F_G = F_T - m \cdot g = m \cdot a \Rightarrow a = \frac{F_T}{m} - g$$

These calculations only work when counteracting gravity. While coasting on an orbit the gravitational acceleration isn't important and thus the TWR may be below one and still work. The acceleration is at minimum directly after launch when the craft is heavy and at maximum immediately before running out of fuel, when the tanks are dry:

$$a_{\min} \approx \frac{F_T}{m_{\text{total}}} - g \text{ and } a_{\max} \approx \frac{F_T}{m_{\text{dry}}} - g$$

The dry mass also includes the fully fuelled upper stages of the craft. To determine the g-force simply divide achieved acceleration by $g_0 = 9.81 \frac{\text{m}}{\text{s}^2}$. As the craft is in free fall, the gravitational acceleration isn't felt by the crew so the accelerations appear to be higher for the crew leading to cancelling out the factor g:

$$\text{g-force}_{\min} \approx \frac{F_T}{m_{\text{total}} \cdot g_0} \text{ and } \text{g-force}_{\max} \approx \frac{F_T}{m_{\text{dry}} \cdot g_0}$$

As the weight of the ship depends on the current gravitation (g) the TWR differs between the celestial bodies.

Conclusion

This guide will hopefully have helped with designing your rockets to allow you to get the job done—whatever it may be—with no test flights first. We hope this guide has been helpful to new and continuing KSP pilots alike.

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This page was last edited on 22 April 2026, at 10:55.